

ActInf ModelStream #001.1: "A Step-by-Step Tutorial on Active Inference"

ModelStream | 1:59:21

hello everyone

welcome to the active inference lab this

is the first active inference model

stream active inference model stream 1.0

and i'm really excited for today's

conversation i'm

daniel friedman and just to introduce

the other participants today

uh ryan go for it uh yep

hi i'm uh ryan uh i'm from the florida

institute of brain research

hi i'm christopher i'm a phd student at

the mrc

cognition brain sciences unit which is

based at the university of cambridge

hi i'm max murphy i just completed my

phd at university of kansas in

bioengineering

uh with a focus on neural engineering

awesome

thank you everyone for participating and

for ryan and christopher

two of the authors of this awesome work

we're going to be exploring

so this is the first in a several part

series that is going to be highlighting

several perspectives and

addressing questions related to the

active inference tutorial paper

of smith et al called a step-by-step

tutorial

on active inference and its applications

to empirical data

so the idea here is for those who are

working with empirical data

to learn about active inference as a method and also for those in the active inference community to be learning about some of the methods that apply active inference if you're listening you're participating and if you have any questions during the live stream feel free to post it in the youtube live chat and we'll try to address it during or after this presentation that we're about to get if you have questions after the live stream please feel free to leave it in a comment form and we'll try to address it and integrate your input in future sessions and to learn more and to participate check out activeinference.org or any of the information in the video's description so that's all the information uh or metadata for this video the way this is gonna work today is we're going to do some introductory questions uh just sort of like asking what in general is this work about what motivated the authors to write the paper the way that they did and then both ryan and christopher are going to share their screens for part of the presentation and they're going to show us a few different things about the work that they've done and then we have a couple of questions prepared on our side but also we're going to be looking at a

live chat if anyone has questions so
just post it

whenever you feel like it in the live
chat and then we'll again try to address
it

so as i stated the intro questions and
then we'll go to the presentations

so first intro question to uh the
authors

is what is this work what is exciting
about it what

motivated you to work on it

um okay so uh so just to kind of

reintroduce myself a little bit more so

i'm uh ryan smith so i'm an investigator

at the laureate institute for brain

research in tulsa oklahoma

and the um the focus of our institute is

um primarily on uh neuroimaging

and sort of neuroscience approaches to
understanding

um psychology and psychiatry with a

focus on sort of treating psychiatric

disorders

um and so for a while now there has been

um the use of simpler computational
models

primarily reinforcement learning models

or drift diffusion models things like

that

um out there for researchers for who are

working with empirical data

who so there are good resources out

there for people to learn those methods

um to apply them to data in their own

research

um however at the moment so active

inference is a sort of much newer field

especially it's sort of a formulation in terms of uh partially observable markov decision processes and there isn't really to date a really clear sort of combined place to learn the sort of practical methods to um to build these sorts of models and um to then sort of apply them to uh task behavior in empirical studies um so the kind of motivation for this paper this tutorial was to allow somebody you know so new students or somebody who's sort of like an early earlier you know junior faculty things like that who wants to go in this direction to give them the resources they need to do that as easily as you know and as thoroughly as possible so even if you start out not really having any background in this stuff the hope was that you know the the first section of the paper would allow you to kind of get enough of a background in theory such that you would understand how to then learn how to build these models specifically for empirical tasks and eventually to learn how to fit those models to uh to data um so that they could be used in your own sort of research um you know even you know for across many different

fields so i would say what's what's
exciting or motivating is
is primarily just that if you're a new
researcher and you're interested in this
but you don't know how to do it
you know the hope is is kind of going to
be like a one-stop shop where if you can
get beginning to end then you'll
know how to do what you need to do to
start using it in your own research
so that's the major driving uh
motivation
awesome i love that one stop shop for a
researcher who wants to apply the
methods but doesn't really know
where to start so that's really a great
idea and
you have version the document many times
to really
address people's input so it's cool how
it's an evolving document as well
so christopher what would you say in
response to those
intro questions um
so i think there are two parts of my
motivation for writing this the first
was the selfish motivation in that i
think
if you want to understand something the
best way to do it is actually to write a
tutorial on it
and so in when ryan asked me to do this
tutorial with him
i was jumped at the opportunity just
because it would be an opportunity to
kind of
actually learn this stuff better myself
and really get into some of the more

technical details

um and so then more broadly speaking

though

i think there are lots if in principle

there's nothing kind of

applying these models in practice is no

more complicated than doing like

model-based reinforcement learning

and yet it's incredibly more common to

see researchers working with

various more sophisticated rl schemes

and i think

if you are a graduate student maybe

starting in psychology or neuroscience

you're like hey i'm

interested in whatever it is uh decision

making under uncertainty

and i mean we have a finite time in this

earth right and a fine amount of time to

actually

dedicate into learning things and so i

think it would be perfectly rational if

it's someone to look at active inference

research

like wow this is super technical i wish

i could learn it but here's this other

thing where i can go on euromatch i can

read sutton and barto

and really totally get up scratch and

use my research fairly easily

and there just wasn't something really

accessible

to help people actually apply in their

own research um so that was kind of my

yeah yeah i mean i should say just as a

little background you know when i wanted

to learn this stuff

originally at the end of my postdoc

before i took my current
faculty position um i the only way to do
it
the only way that i could learn it was
actually to go you know hang out at the
phil
in london with carl for about four
months um
and you know just ask people a ton of
questions and
sit there and that was the only way you
know so without having to
you know physically go to uh one
location in the world i was very far
away from you know my current
institution
and without doing that there really was
no way for me to do it so i mean this is
sort of a out of empathy for my past
self
that you know this is the sort of thing
i wish i would have had available to me
so i could learn this stuff
independently
cool so max um what is your
background or what got you really
excited to work through the paper and
develop a lot of the examples out on
your own um computer and everything so
what what got you excited or where are
you coming from today
sure um yeah so uh maybe i'm kind of
your target demographic in some sense
uh i don't know that i could really be
considered a junior investigator
although i hope i could someday be that
um and i'm interested in um i'm much
more on the

the motor system side of neuroscience as opposed to the decision making so this just the whole um lexicon it's a little different but i am seeing so many commonalities when i'm coming from a signal processing background i think of things um in terms of kalman formulation and trying to integrate sensory and motor information i see a lot of similarities there and so for somebody like myself then looking at some of fristen's work and seeing how he's described you know the anatomy of uh inference and could this apply also in motor systems i'm very interested to see how this might apply in my own future research and work cool so let's just say that you were speaking ryan first and then christopher to that early investigator researcher of any age who is like okay i get it um in reinforcement learning it's about reward or reward-centric learning is about reward what is active inference how is it taking a different perspective on what it is that organisms are doing than that kind of classical account of reward based learning for example or whatever you'd like to contrast it with but what is the bridge from the kind of

assumptions

and implications that lead somebody to
work on that classical framing
versus what is the um difference in
thinking that is active inference and
how is that manifested in the model
so i think this is actually a really
important thing to bring up right from
the start

is that so there's a there's a kind of
big distinction between
um when people talk about the free
energy principle broadly or the
philosophy of the free energy principle
and there's a very kind of uh you know
somewhat wide chasm between that
and what gets called active inference
nowadays where active inference is a
corollary

of the free energy principle but um
you know active inference as formulated
in terms of partially observable
partially observable markov decision
processes is a
quite a bit narrower right it's a very
specific sort of
discrete state-space generative model um
and that has particular sorts of
elements to it um
and it it doesn't actually require
knowing a lot of the things that people
talk about with respect to the free
energy principle more broadly
um it appeals to um free energy and
expected free energy
as functions right as more or less cost
functions for
figuring out what the best choice to

make is um
and it but but at the end of the day i
mean when you apply
these sorts of models to behavioral
tasks that they used in studies
it's actually there's quite a bit of
similarity i mean you can really see
active inference models or at least the
mdp formulation
as as just a particular is kind of like
what chris said is just a particular
kind of flavor almost of model based
reinforcement learning
um you know so instead of i mean there's
there's some sort of technical
differences
right where um the way that sort of carl
and them have set it up
what you're doing is kind of like it
allows for kind of a fully unified
bayesian way of doing
reinforcement learning and decision
making processes
and you do that by instead of calling
something
a reward per se you define this
probability distribution
over observations that gets called a
preference distribution
and so reward then becomes a kind of
probabilistic preference
to observe some things over others um
you know so if you call
a particular observation a reward um
then calling it a reward just amounts to
having a
precise preference distribution that has
a precise

high value over whatever that rewarding outcome is

and then the agent is simply driven to make decisions that it thinks is most likely

to get it to observe the thing it prefers

right so this preference distribution is more or less a way of specifying what is rewarding to the agent right what the agent is seeking

um one thing that is kind of nice about active inference that's a little different

from reinforcement learning per se

or there's a few things i should say one is that um

making decisions based on trying to minimize expected free energy which chris will cover

doesn't just try to maximize reward

it also simultaneously tries to maximize information gain

so so what will happen is you know say

say an agent starts out in a dark room

you know just to take a very kind of contentious

you know philosophy of the fep sort of example

right say an agent starts out in a dark room um

which means that it doesn't know where anything is

right so that means it has a lot of uncertainty over what state it's in

um and let's say simultaneously in another room

there's a fridge with some food in it um

then there will be two different sorts
of drives there right there'll be this
epistemic drive
to do whatever it thinks is gonna give
it the most information gain
right which will be go turn on the light
right so then it'll know
what state it's in right whether it's in
room a or room b
right or whether there's a couch in the
room or whether there's a
tv in the room or whatever right um and
that's not
reward per se that's just trying to make
the app
choose the action that's going to
maximize the amount of information you
get
the other the other chunk of it though
is to maximize
observing preferred outcomes and so the
agent will also be driven to just leave
the room
to go to the fridge you know to observe
itself eating food um
you know so in practice what will happen
that typically won't happen
at least in a standard reinforcement
learning setting
is that the agent won't just try to
maximize reward directly
a lot of times it will first choose the
action that will
help it figure out where it is so that
it's more confident what to do
to get the reward later so
so it's so it's really not again in
practice these models are

are just a nice fully bayesian way to
integrate

perception learning and decision making
where decision making is driven to both
maximize information gain and maximize
reward

christopher thanks a lot anything else
to add there

um no not particularly i i think the
only thing i would say and i'm going to
say this in a little bit

on a slide is if you're interested in
the technical details what ryan just
said there are two really good papers
um maybe we can make these slides
available afterwards as well

um one's by norsajid and i'm not sure if
i'm sorry if i'm

butchered your name there um the other
one's lance lot decosta

and they have two excellent papers
comparing it to reinforcement learning
so

formulations excellent um just to draw
out a couple points

ryan from what you said so one is that
uh

the active inference framework and
specifically the

po mdp the partially observable markov
decision process

instantiation of active inference is a
corollary

or like a derivative of the free energy
principle

and there's a lot of philosophy and
contentiousness around different aspects
of that we're going to read a paper in

active inference stream 14 by mel
andrews about
the math is not the territory about the
philosophy and the status so yes if you
want to go
that way there's a whole rabbit hole
there but this is kind of like a tool
like a linear regression
you don't get caught up in the weeds on
number theory or on the information
space that it's about
where the calculation is useful and
tractable
it's going to be a tool so that's kind
of where we're coming at at least today
and in this series another interesting
thing
and in contrast with reinforcement
learning which is kind of like
uh reinforce what works and don't
reinforce what doesn't work that neurons
that fire together wire together
positive reinforcement schemes there is
a reward preference built into that kind
of model
but when there's a basin of low reward
it's often difficult for those models to
latch on
so we saw that in the paper of alec
chance at all
in active inference stream eight scaling
active inference
that kind of showed how even up against
these very very
large scale machine learning models like
q reinforcement learning and things like
that
other state-of-the-art deep learning

models that the active inference
trolley car and control theory robots
were able to work really well because
first they did what you just described
ryan which was
they kind of went into an explore mode
before going into a more fine-tuning
mode and so in doing so they transcended
the explore
exploit simple trade-off it's not just a
knob in this model
it's not a coefficient to balance
explore versus exploit i hope will draw
out an understanding that they're
actually related in a different way
because there's something that's model
based and generative that's happening
and so
really a ton of interesting stuff so at
this point
i would think we could go into the
presentations
if people have any questions that arise
put them in the youtube chat and i'll be
copying those out
for addressing them later and um other
than that let's just work through the
presentations however
ryan and christopher have set up and
preferred
cool okay go ahead and just share your
screen chris you can just go ahead
yeah it's like just kidding
okay so i'm using a dual monitor setup
we'll see how this goes
um
okay sorry i now have to open system
preferences sorry

to allow microsoft teams to use it sorry
about that
um
okay should be with you momentarily
cool
good to digest really it's really
it's interesting stuff and we're all
learning by doing
here so i just really appreciate and
have just seen a lot of
appreciation for this kind of work
because in some ways it's like the
missing
piece between a lot of these
hypothetical or abstract or scaffold
models
or what people have heard about with
respect to what active inference is
and then how it's deployed how it's
enacted so to take the inactivist
insight seriously as practitioners of
science
and as communicators as well it means
making this kind of work and
showing us how you do it as we all are
trying to like
figure out every level from literally
the tech to
the best way to make it rigorous and
accessible
the best ways to talk about it so yeah
what what do you think about that ryan
um yeah i mean i totally agree hold on
to chris
yeah actually i've never used um this
there should be a little thing in the
upright corner it's like an arrow
pointing into us into like a square

share content

are you guys both on pcs yeah

yeah okay so yeah so

it should be upright corner like right

next to the leave button

okay um or ryan you could present

first or yeah um

i actually have a question i i have a

question it will really work for me to

present first i think i think

you know yeah what i'm gonna do not

gonna be so comprehensible before chris

got it i actually have a question that

will take a couple of minutes uh

christopher while you're figuring it out

is

what tools would somebody need to follow

along

so what programming language or what

interface

do they need what background knowledge

do they

need is there a software to download is

there a course

what are the prerequisites for doing

what we're about to basically jump into

um yeah so i mean i'll just say that

that you know a lot of this stuff was

originally designed

by carl fristen um and if you know

a lot about carl um you know before he

proposed a lot of this active inference

stuff you know he

was uh sort of most uh you know famous

or his his biggest contribution was

putting together

uh sp spm within matlab spm is um

like a it's called statistical

parametric mapping software but it was more or less the original way of doing fmri so like analyzing functional and like neuroimaging data and so a lot of the scripts and dynamic puzzle modeling i should say which is a particular sort of approach to doing your imaging i mean so a lot of basically all the standard resources all the standard coding routines everything for actually building active inference models right now is primarily an spm which runs within matlab um so and all the kind of supplementary tutorial code that we've provided we've provided i think six six or seven different um different supplementary scripts where you can actually run the simulations and build these models yourself they are they are all they're all in matlab um so if you so if you have matlab then you need to have downloaded spm um and with matlab and spm you um you will have everything you need to do to open our supplementary code and to uh and to follow along so a lot of the tutorial is actually set up so it assumes that you're kind of have the paper open side by side with the code so you can actually click on you know different options to simulate and reproduce all the figures in the in

the tutorial paper

so it's uh yeah so the short answer is

you need matlab and you need spm

um thanks for that and you need to have

some minimal

ability to you know work with work with

matlab

just just to piggyback on to that for

anybody's

benefit watching this in the future

whenever you're watching this uh

once you get the spm folder you know

you'll unzip that you'll get that folder

you'll put that somewhere on your

computer

you'll want to make sure that you add

path from your matlab workspace when

you're in the supplementary code

provided

by uh ryan and christopher

when you're in that workspace add path

of that folder wherever you put it

add path of that folder slash

toolbox slash dem

and then you should have access to the

additional spm12

scripts that you need in order to make

use of the functions they've provided

yeah no thank you very much i mean

that's that yeah for someone who's never

used matlab yes there's a few

steps like that which are super

important um

yeah because basically all these scripts

are actually within yeah within

the dem toolbox um

of spm um and uh all the scripts they've

provided

um call on uh sort of
sub functions um within that are within
spm
um so so yeah it won't unfortunately
uh none of them will work without
without spm at the moment
um although there are certainly efforts
to try to make
um some of these routines more sort of
generally accessible
and like free software um like alec alec
chance for example
is in the process of um i think writing
a uh like a python
version of the um of the main um
active inference uh mdp script which is
called the
it's called uh uh spm
underscore mdp underscore vb underscore
x
and that just corresponds that just mdp
underscore vb underscore x corresponds
to markov decision process
underscore variational base underscore x
which stands for factorized
um and i'll i'll go over what
factorization is
um and how you actually code that in
once we get to
my section um
okay i'm gonna ask a follow-up there so
i have my
desktop computer i have matlab and it's
all working with referencing
spm most updated version
and i've run through the examples that
we're going to be working through maybe
today or the ones that are in the paper

so i kind of hit enter and i got things
to work
everything's all the scripts are working
fine
what information from my system do i
need to know
like what measurements are going to be
relevant for me to bring to the table
if i'm going to do a t-test for whether
two ant colonies are
a different uh size you know i know what
kind of machinery i'm going to need you
know two columns
in the excel spreadsheet or something
but what kind of data
or what attributes of the system are
important
to bring to the table for an
investigator to utilize
the examples that you're providing
um yeah so i mean so i mean there's a
couple things i mean so
so first the um
you know so the first goal is you know
you take a particular
behavioral task and you have to figure
out
what the right generative model is um
for that task
um and this is the sort of thing that
i'll give an example of i can actually
show you multiple examples of this
um and so once you have this or
generative model set up
then you can run simulations and those
simulations will generate
observations and they'll generate
expected actions um

so so what you need then is you need
data
from an actual participant performing
that task where you'll know
on each trial um what the person
observed right what the tasks amelia
were
and you'll know what action they
actually chose
and so then what you'll do ultimately is
you'll do what's called parameter
estimation
which is and again this will be a lot
more comprehensible
once i actually show you but but
basically what you try to do
is you try to find the set of parameter
values in the model
that generates behavior that's most
similar to what the participant actually
did
and so once you find the parameters you
know say one has a value of two and the
other has a value of four
and that generates behavior that's
identical to whatever a given
participant did
then those values of 2 and 4 you can use
those
as individual difference estimates you
know so if you have those values for a
bunch of different people who behaved in
different ways on the task
then you could say hey you know is there
something different right about the
people that have a parameter value of
four
versus a parameter value of eight right

can i use that to predict for example
you know how well somebody's going to
respond to a particular treatment
in in computational psychiatry um
um hi chris twice yes

um

yeah but so but so but so that's that's
kind of the idea

um you know and i mean i can't i can
share my screen in the meantime

um i don't know well here i can i can
perhaps go through

some of this um so i'll just share my
screen until

i'm good yeah sorry that was this whole
thing was

um oh you're good okay

i think i am at least can you guys see
my screen uh yes perfect

uh right here okay great sorry about
that there was some a whole bunch of
privacy issues

um they had to enjoy the passwords there
anyway

okay we are here so

okay so are you guys seeing my present
of you or are you seeing the full thing

looks uh we see the we see the full
slide yeah just the slope awesome

okay so this is just kind of part one um
after i've done

gone through this uh ryan's going to
take over and go through some of the
practical aspects of it

um and so then just kind of reiterate
the scope and purpose

our target audience really is kind of
researchers in neuroscience and

psychology that don't have a strong
kind of quantitative background in maths
or ml in particular early career
researchers
and we just really want to provide
people with a requisite background
to actually apply this in the context of
their own
okay and then so just to quickly
highlight some really fantastic other
resources
um the first paper is by lance lot de
costa which is an incredible technical
review
just came out mathematical psychology um
north has a really
other incredible paper this came out
neural computation
and i think the um comparison to
dynamic programming and bellman
formulations um is still in pre-print if
that's right ryan um but as well there's
some really phenomenal
informal tutorials with oleg solo shock
i'm sorry if i butchered your name but
anyway they're
fantastic kind of informal tutorials on
medium
and then lastly is kind of the closest
thing to what we're doing today
is these lectures by philip
schwartenbeck
in the computational psychiatry
summer school and these really are
fantastic
the only difference is that they work
with to my knowledge i think they work
with the unfactorized mdp scheme which

is a little less flexible than what
we're working with
um okay and so
there are a lot of ways of kind of
motivating active inference
i think kind of the way that's most
intuitive to a lot of people in
cognitive science or backgrounds in
cognitive sciences to
kind of start from perspective of the
bayesian brain and so
interrupting if anything's kind of
moving too quickly or too slowly or
anything like that but broadly speaking
the idea is
is that the brain encodes a generative
model of the environment
where this generative model is just kind
of a joint probability distribution
and then outside or in the outside world
as it were
there are states and observation states
give rise to observations
and we use the generative model in
combination with bayesian updating
to infer the hidden causes of our
sensations
from the observations and then we take
actions based upon
our kind of internal model of the world
that couples us back
to the generative process that if that
makes sense and so there's this kind of
perception
action loop that's always going on so
chris i think i think you might want to
tell people just what the p o comma
so that is a joint probability

distribution over observations
states and policies where policies are
just actions or sequences of actions
um that are available kind of all of the
actions that
are afforded to the agent essentially um
okay and so then kind of
just we start with like a re illustrate
this we'll just give like a really
really basic example
and so you might be imagine being
presented with
kind of a shadowy shape like this and
then you might and what you want to do
is infer the causes of that shape based
just based upon
the observations but you don't just have
observations right
you have some prior knowledge about the
world
um and so we can specify that here so we
might say
where two possible causes in this very
limited example
there is we might say that the shadow is
caused by a convex surface or a concave
surface
and they have fairly similar prior
probabilities however we also have the
structural prior that we've kind of
acquired through a lifetime experience
of just light emanating from above
and under the kind of a structural pry
that light emanates from above
the likelihood of the observation of
that particular shadow
kind of conditional being concave is
much much higher

okay and so then we can then combine the two the prior and the likelihood to give us a joint distribution where this is just kind of conditional on a specific observation

we can then and then we sum over the states in that likelihood to give us our marginal likelihood and then divide our joint distribution or our generative model by our

marginal likelihood and that's called model inversion or is often

colloquially called model inversion

and from there we get our posterior distribution this is the probability of states

conditional on observations and so we started with a prior

and a likelihood and got to a posterior through bayes rule and that is kind of formally speaking the optimal way to infer

the probability of hidden causes given observations

and the complication to all this is that the marginal likelihood is generally speaking computationally intractable

um so i in the case kind of these very simple discrete distributions

the number of sums that you have to perform scales exponentially with the size of your hypothesis space so that's extremely impractical um and

kind of more realistically when you're working with in continuous state spaces if you have non-gaussian or non-linear um signals

they're just analytically the

distributions is analytically

interactable

so chris i think i think uh max had a

question yeah yeah sorry

yeah real quick uh just just to really

explicitly uh touch on

the idea of hidden hidden states and

hidden variables

in this particular context uh would it

be fair to say that it's not only

the convexity or concavity of the the

potential

you know thing that i'm looking at but

also i could think of

the light source the direction from

where the light source is as being

another hidden variable

and those things together would it be

fair to term those as under my

markov blanket or is that uh getting a

little too far out from where we want to

be talking right now

yeah so markov blankets don't really the

kind of markup blanket concept here

doesn't really

come in in in any really interesting way

i mean you can formulate it that way but

really the idea is just

you have this prior expectation so when

you're looking at the little gray disc

right there under observation

most people when you look at it even

though it's actually just a flat

shape with some you know it's just a

flat two-dimensional thing that has

a little bit of darkness in one place a

little bit of lightness in another

um you know if most people see that as

being concave
even though it's a flat 2d thing
right so the idea is is just that
there's a reason
right why people typically see it as
being concave as it's kind of like
popping in as opposed to out
and that's specifically because there's
this prior belief
that um that light sources tend to come
from above um so the the hidden states
here
are kind of there is a joint in the
likelihood which just means
that there's a in the upright where it
says likelihood there's a concave comma
light from above right so those are two
different hidden states
what it's saying is um if
the hidden states are concave and light
from above
if the combination of those things is
what's generating
the what i'm seeing with that gray disc
thing
um the probability of the shadow shape
that the shadow pattern that you see um
is 0.9
whereas if it was convex and light from
above if those were the two hidden
states
then the probability you get that same
that same shadow pattern is only 0.1
right in other words it'd be really hard
to get that
pattern of shading if if the thing was
popping out
under the assumption that light's coming

from above

um so so the observation here is shadow

and the hidden states are

concave and light from a concave or

convex and light from above and like

from below

okay thanks yeah just one other quick

example there would be like the chess

board

with a shadow and a dark square it's

perceived under a deep cultural prior

that chessboards are regular

alternating grids it makes an ambiguous

shaded square appear as something that's

quite different than it is people can

look that up

and then another uh thing this reminds

me of is like using a t-test

to test whether group a versus b is

taller or something like that

it assumes but um it can be an

assumption that can be

slightly bent is it assumes that other

than the variable being considered

other things are sort of fixed or not

matter otherwise your statistical

test is basically misleading

because it could be capturing some

totally confounding variable

in the framework you put it in and so

whenever you're talking about a real

biological organism you're talking about

conditioning on certain things

and then given what is totally

conditioned out of the picture

so i i kind of see where you're going

with the blanket but it's

not specifically in this place but a

good question

but it's like depending on the
irrelevancies that we can condition out
then we're looking at the conditional
relationships between different
observed states versus the hidden things
so

cool stuff and again in the live chat
people can post any
questions on it continue christopher
yeah thank you

um yeah thanks for covering that ryan um
okay so then the idea is is that the
marginal likelihood is generally
computationally intractable

but kind of borrowing some ideas from
statistical physics

what you can do is actually use some
approximation techniques and instead of
evaluating the marginal likelihood
directly

we can evaluate something that's always
provably greater than or equal to
the marginal likelihood i'm so going
through that

right now okay and so we're just going
to take the logs kind of for
mathematical convenience and the reason
we always work with logs or generally
speaking is because

log log algebra is just a lot easier
just because it turns multiplications
into arithmetic

um so things it's just easier to work
with essentially

so we'll take the negative log of our
marginal likelihood

um this is also sometimes called

bayesian model evidence and
when we have a negative log of a
probability it's also called surprise
and information theory which is kind of
the terminology i'm going to use the
rest of this kind of presentation
and so you can see on the right-hand
side of the equality this
is kind of the sum rule of probability
we can get back our surprise by just
summing over
kind of all of the states under our uh
joint distribution over states and
outcomes
and then we can kind of do a little bit
of a trick here so what i'm going to do
now
or show you is that we are going to
multiply this joint distribution or
generative model by some arbitrary
distribution
we multiply and divide by the same
distribution
and so nothing has changed here this
quality still holds
i could cancel these if we wanted to but
i don't want to
and that's in particular because i want
to take advantage of something called
jensen's inequality and that is that a
result showing that the
expectation of a logarithm is always
less than or equal to the logarithm of
an expectation
and so the idea is is that this sum
here where the logarithm is inside this
summation
where sorry so we're taking we're

summing over
this kind of difference between a
generative model
and this approximate or target
distribution
when the logarithm is inside that it is
always going to be and we're in kind of
negative territory
it's always going to be greater than
this quantity on the left-hand side
where this left-hand quantity is equal
to surprise because
the log is outside the sum we can cancel
these kind of approximate distributions
right and end up
back at surprise on the right hand side
we can't do that
this right hand side will only be equal
to surprise
when our generative distribution
perfectly matches our approximate
distribution
and so this quantity on the right hand
side here
this is variational free energy and so
the idea is
is that we find the probability
distribution which we make simplifying
assumptions about and so we can
kind of evaluate analytically
we make simplifying we find the value of
this distribution
that best minimizes f
where f is variational free energy does
that kind of make sense
hey chris just one thing are you like
trying to like move your mouse around to

point to things

no okay i'm actually putting my mouse

i'm scribbling my mouse a little bit on

the live stream they should be able to

see that

but christopher thank you for that

really awesome

example because it really clarified a

few things

any other notes to add here otherwise

this is cool to continue

yeah okay so then just kind of give a

really toy example of this

so we can define our approximate

posterior

kind of arbitrarily is a flat

distribution we'll have a true posterior

generally speaking we don't know what

this is but for illustration's sake

we're going to give

kind of give it to you and then we can

have a joint distribution

and an observation what the observation

will do is it's just going to select

a kind of a column from this joint

distribution

and so then we have we can now enter

this into the equation we saw on the

last slide

and we'll slowly but surely in each

update

nudge our posterior distribution

such that it minimizes f on each step

and at the third step you see what we

should see is that when f is at a

minimum

it is equal to surprise and when f is at

a minimum as well

it's at a minimum because the true
posterior
and our approximate posterior match
so that kind of makes sense um and so
the idea is is that
by performing bayesian inference and
having and kind of forcing
some arbitrary distribution to
approximate our posterior
we come up with an upper bound on
surprise or we come to
an upper bound on surprise let me kind
of just
walk through and just double check here
so q
of s is our estimate of whether the coin
is a fair coin 50 50 means that it's 50
50 and then 80 20 is the reality
on the table so to speak so it's almost
like
first we click from 50 50 we see heads
come up
and that contributes a little bit of
evidence that maybe heads is more likely
so we go from 50 50 to 60 40.
and then something happens again and we
click on the second update
to 7.7.3 we click eventually to 0.8.2
and then if we click all the way to
point nine point one
so again in 90 10 all of a sudden we go
uh back up from 0.693
back up so if we were to plot this final
estimated f we'd find that we're like
going downhill getting better
get to 0.693 in this discrete 0.1
movement space and then we pop up a
little bit too far we overshoot

and we get a little bit more surprised
by the distributions that we're
drawing yeah exactly that's perfect um
well the only the only thing that i
would say
that's i mean a little different just to
note is that there's nothing in this
example where you observe something over
and over again
this is this is just a single
observation once
and you're trying to figure out what
what is the you're trying to figure out
what the true posterior
is trying to get as close to it as
possible so all you're doing is saying
here's the single observation
and i'm going to try out a bunch of
different values for q_s
a bunch of different values for that
approximate distribution
and i'm just going to find the one you
know through this kind of iterative
updating thing where i slowly move q_s
you know to different values you just do
that for a single observation
you update until you find the minimum
free energy value which just tells you
that your approximate posterior the q_s
thing is as close
is really close to the true posterior
that you couldn't figure out
on your own because the problem was too
tough to solve with exact bayesian
inference
um so this this is kind of corresponds
to what people might talk about as
prediction error minimization

right so you just see this observation
once
and the brain tries to minimize
prediction error by minimizing f
by just moving beliefs until it finds
the
the belief that minimizes f
which is the same thing as minimizing
prediction error
um look at that if that makes sense
let me clarify because i i think that
really helped me understand it and
at first i thought wait with one
observation because at first when i
described it i was thinking update was
tied to a new observation from the coin
but then i thought well if you're only
getting one observation the maximum
likelihood model is a coin that only
comes up heads
because you only have one observation
but it's not one-shot
parameter learning naive it's actually
a a tethered estimate that's tethered
even loosely but non-zero tethering to a
prior
0.5.5 some people will say uninformative
but all priors are informative
they're all what they are and then it's
almost like because of the logs
even though you only observed the coin
come up once it's like all right it's a
little too far to update
to 99 one just from seeing one coin flip
i would need extraordinary evidence for
extraordinary
updates and so it's like that one heads
observation

updates you from 50 50 to near
80 20 which does happen to be close
to the actual probability of the coin
and it also happens to navigate this
explore exploit in an interesting way
because it updates it but not all the
way so it's just kind of showing
how the bayesian updating brings some of
that
wisdom of multi-observation learning
like slow updating of parameters and
uh sequential uh updating to
a little bit of a different context so
yeah i mean
i do want to clarify here a little bit i
mean there's there's no actions here
so there can't really be explore exploit
right you can't
you can't choose to look over here
versus over there to gain information
right so explore exploit is you know
specifically in the realm of making
choosing actions that will minimize
uncertainty
versus maximize reward um this is this
is really more just
with with normal free energy right we're
not too expected free energy yet which
is the decision-making part
with normal free energy there's you can
think the simplest way to think about
um free energy is just in terms of
um complexity minus accuracy
which is equivalently complexity plus
prediction error
and all that all that ultimately means
is so the complexity thing is basically
how much you have to change your beliefs

um so it's basically saying what's the minimum change in my beliefs that will make my new beliefs as accurate as possible right so i have to move my beliefs as little as possible while also minimizing prediction error if that if that makes sense so this what this generative model is saying is just that the probability that we're in state one is point eight if i were to chew if if i were to get this observation where the probability of being in state two is only point two where to get this observation right so so i mean the probably a better example than the um than the like heads and tails thing would just be would just be something like the concave complex example we gave before right there is some possibility where where light from above you know where light from below could cause the shading pattern but it's just much less likely right so you're just trying to find the the belief right that it's convex and light from above that that is the one that most likely generated what you're what you're seeing and it might take a bunch of updates like that given just what you see once to kind of arrive at the best fitting belief there's a related question in the live chat that i'm just gonna ask because it's on the topic

they asked is this approximate bayesian inference thing

called something else in stats outside of active inference or is this unique to active inference because this isn't sequential bayesian updating as you mentioned

this isn't a standard bayesian filter

what is this called outside of the active inference field

so yeah it's just variational bays variational bays where does the partially observable markov decision process come into play

uh we'll cover that in a moment's time actually yeah it doesn't

it doesn't yeah it doesn't come in yet

so i think one thing to say is kind of writing this tutorial it's hard to

please everyone right like in going through some of the feedback like we get we got some feedback where people were just like utterly confused

and wanting clear explanations and then at the other end of the spectrum we wanted there were people wanting much more technical details like how does this relate to things like gradient descent

because this is just kind of a very simplified cartoon example of something called grading of

a gradient descent scheme where you're doing a gradient descent

on free energy um and so i would just say kind of the flag

of these issues is one i would say if something's unclear in this

presentation read the paper because we
gave things
we cover things a lot more detail in the
paper and if that isn't technical enough
for you
actually go and just look at our code we
supply
a kind of standalone script where we
it's extreme and it's extremely well
commented and from there you should be
able to figure out everything that's
going on
so all of this stuff in this
presentation is kind of necessarily
simplified because you don't
really need those technical details to
kind of start
using the framework or start getting
intuition for how things work
but once you do have an intuition and
you want more just go and see the code
i would say um awesome
it's like you can use the anova package
in our
without going into the source code it's
helpful it's a tool for scientists
and then if you're curious about the
underpinnings of statistics
and perennial philosophical debates
there's a literature
and a search bar but today it's about
the applications of these methods
which is awesome so thanks for everyone
for the questions keep them coming but
this is great
discussion i really appreciate it
okay and so just kind of to recap there
i haven't covered action yet i'm going

to cover action a little bit
but just kind of to recap the idea is
under active inference
organisms are kind of we model them as
if
their phenotype is in their body and
their brain kind of embodies a
generative model of the environment
and organisms kind of invert a generated
model to arrive in
approximate posterior distribution over
the hidden causes of sensory input
and they do this by minimizing
variational free energy
so that should that much should
hopefully be clear by this point
and so then on to the generative model
and it is a very specific type of
generative model
namely it's a partially observable
markov decision process
and so there's kind of a graphical
representation of this as a bayesian
network
on the right-hand side and i'm not going
to kind of give give too much detail
about this
right now i will actually build it up
step by step
but just kind of prelude the idea with
pomdps
is that they describe transitions among
hidden unobservable variables
and the sensory data that's generated by
the variables whoops
so those arrows mapping from states
to observations are kind of give
information about the direction of

influence or conditional independence
conditional influence so kind of the
arrow between the purple
node o and the green node s
is mediated by the a matrix which is a
likelihood
and the transitions between states are
mediated by a b
matrix which is a transition probability
and so the goal of active inference with
pomdps
is to infer states and action sequences
or policies
by minimizing various forms of
variational free energy
okay so start with a really really
simple example
we have static inference and the idea
with static inference
is this this is just a graphical
representation of
bayes rule essentially we have a
prior which is encoded in our d vector
we have a likelihood
encoded in our a matrix and we end up
with an update equation
which is a soft max function a softmax
function is a normalized exponential
whoops
and so i'm not going to kind of run
through that explanation
in detail i'm just going to kind of
leave the slide here so people can
figure it out for themselves and pause
and go back and convince themselves that
this is true
but for this very simple example uh the
active the inference scheme that we're

using here is formally equivalent it's just an exact inference scheme so moving into dynamic models specifically so where states change over time these are also called hidden markov models this is when we have to start making approximate or when things are no longer equivalent to a prop using active inference are no longer equivalent to exact inference and so here we have our transition probabilities encoded in a b matrix and this is just essentially like the probability of some state at t plus one conditioned on the previous state and so you can see over here in terms of the update equations we're now in we're in log space and this little sigma thing here is a softmax function which just normalizes this equation and the idea is is that this is kind of sorry my screen keeps flipping um anyway uh the combination of d and b so our two priors in addition to our likelihood will give us the approximate posterior and the reason we have this one half in front is just because in practice the approximation scheme that's used by active inference namely variational message passing tends to overestimate

kind of the value of the posterior and
so this is kind of just a way of
compensating for that and so anyone's
kind of interested in technical details
there

see thomas parr has a really excellent
paper out in scientific reports on
neuronal message passing schemes
under various approximations to free
energy

okay and so then the form of the free
energy is just down here

um and so the idea is is that by

iteratively applying these

updates we minimize free energy

and you iteratively kind of every you

start you do a full round of updating

every time you get a new

observation

okay but what about policy selection so

the idea with policy selection is

roughly speaking

that policies are just state transitions

that the agent has control over

so imagine you are a psychophysics

experimenter

and you have just a very simple boring

example of someone kind of

uh say estimating the orientation of

some hard to see stimulus

and so the policy space there is tiny

it's two options it's left or right

let's say

or you could have an agent that you

could be simulating something more

interesting in psychophysics

you could be simulating an agent

navigating a maze and then the policy

space is much larger they could go
forward they could move forwards
backwards
left right et cetera et cetera
and so the idea is is that active
inference agents by definition
need to select policies that will
minimize variational free energy
but that relies upon
observations that have yet to come so
that's kind of a problem
but the way around this is to treat
observations as random variables
and then what you do is you minimize an
approximation not to surprise which is
variational for energy but to expected
surprise
and that approximation to expected
surprise
is expected free energy and so this
expected free energy has two
key components the first is the expected
cost
and this kind of minimize the idea here
is to minimize expected cost
you need to minimize the deviation
between our predicted and our preferred
outcomes
and so this is kind of what we were
talking about ryan was talking about
before so this c vector
encodes a distribution over an agent's
preferences
so for example policies that minimize
expected costs
are policies that kind of bring about
observations that the agent prefers
for example so if i prefer to kind of

have
my body temperature within a certain
range the policy would be
maybe staying inside because i live in
england and it's like negative one
outside
um for example now
expected ambiguity is a little bit this
this is kind of the epistemic drive or
the information gain term
this is kind of the expected entropy of
our likelihood distribution
so the idea here is that to minimize
ambiguity
you will have an a matrix right an a
matrix if
the idea is to minimize kind of
ambiguity you need to select
observations in that a matrix that
are maximally precise and so to return
to
or take actions even that will make
those things that a matrix maximally
precise
and so if i'm in a dark room to give an
example before the thing that will make
or select kind of met or policies that
will make that distribution maximally
precise
are policies like turning light on
and so then to minimize free energy as a
whole you have to minimize both of these
things
and you can do this in terms of one-step
policies just looking one step ahead
or you can do it look in using deep
policies looking many many steps ahead
hopefully that's pretty clear is there

anything you'd like to add to that ryan
or
daniel anyone else yeah i mean i guess
i'll i guess i'll just say i mean to
just to make it kind of as
as clear and explicit as possible for
people who don't have a background in
you know and what this whole thing you
know what the seclusion
means right so the idea is just if you
look at that term above expected cost
that's called a kl divergence and so
basically it's just a value
that encodes the difference between two
distributions or the dissimilarity
between two distributions
and so that first one that q_o given p_i
that's just saying what observations do
i expect
given that i choose to do this versus
that um
then that second one that p_o of o that's
the preferences
so basically what it's trying to do is
just minimize the difference between
your your preferred observations and the
observations that you expect
given that you choose to do you know
thing one versus thing two
um so it's really just choosing the
thing that's going to get you as close
to that you think is going to get you as
close to what you want as possible
it's very just you know reward seeking
more or less
and that second one the expected
ambiguity that each thing
stands for entropy and basically entropy

is just
the higher the entropy the flatter a
distribution is
so think of a distribution that's like
one is like 0.5.5 and the other one's
like 0.8.2
if you chose the state that would
generate 0.5.5
over observations then it wouldn't tell
you anything
because either observation you got it'd
tell you
there's a 0.5 probability that you're in
one state and a 0.5 probability you're
in the other state
right whereas the other state would
generate a 0.8 or a 0.2
then that one will give you a lot more
information because you observe if you
observe the thing that indicates 0.8
then you're really confident what state
you're in
and if or if you observe point two
you're really confident you're not in
that state
right so you're just seeking out the
thing
that you think will get you what you
want as much as possible
but also moving to the states that are
going to
give you the observation that's going to
tell you the most about where you are
if that if that if that makes sense so
just again just for people who
who aren't necessarily as familiar with
reading the the kind of
notation in these equations thank you

max

um yeah just one one point of clarification and then i just wanted to tie it back to what we the example we discussed previously

so first i want to just make sure i'm understanding correctly that the d in that equation is the the kl

uh the diverge so that's the that's an operator it's not the same as the d that's in the figure uh in the block diagram well that's

that's the kl divergence that's a yeah that's kind of a standard way of representing it in other papers that's just kind of paste it in but

um it's not very clear that should be the kl it has nothing to do with d d sub n the d that's in this graphic in the block diagram would that be analogous in our previous example where we were talking it when when we illustrated step by step that would be our

0.5 0.5 but it in this context could be much more complicated than that yeah in a simple example yeah it would just be your your prior

right so if ahead of time you think that like

ice cream is more likely than donuts right

then that could just be like 0.8.2 you know that's

or if you have no idea whether it's going to be ice cream or donuts it could be 0.5.5

and for and for anyone who is uh

like wanting to get into optimization
and how do we know
whether or not this is going to converge
when we do our message passage
our message passing algorithm back and
forth there was a
really good citation and that's the one
that was alluded to
at the beginning of this uh on the
slides right the technical papers is
that correct
uh so the citation to do with message
passing in particular is a paper by
thomas parr
so that is in scientific reports um
yeah we can i i think do you have show
notes i i assume we can yes
please i have a couple other general
questions from the
live chat but i think we'll take them at
the end of your presentation
as we turn towards some of the
neurobiology beginnings and
um a few other aspects so continue
christopher yeah cool thank you
um okay and so then kind of just to
close or just to briefly recap
so there are multiple stages to this to
policy selection
so under a generative model that has
multiple policies
you need to minimize you minimize your
free energy with respect to each policy
your variational free energy and so you
might think about this as
say you're coming to a set of traffic
lights and
you have a whole bunch you could turn

left right or go straight ahead
those actions are possible but you then
get
sensory input that says there's a no
left turn sign
so that would make that policy give an
extremely high free energy value
and so that would eliminate it from the
plausible policies that you can kind of
evaluate right
and so then expected fringe is what we
just talked about
and the posterior distribution is
actually
uh the net soft max function over both
of these and so this because
kind of this is these are both negatives
um
policies that best minimize both
variational free energy and expected
free energy
will have the highest posterior value
um and then kind of is that is that
clear there are a couple of caveats to
all of this
um but just wanting to make sure ryan do
you want to add a thing here
well i just say just just so anybody who
doesn't know like what a
what a soft max function is yeah so so
all that
all that means is that like for instance
when you take
 $-\mathbb{E}[\log p(y)] - \mathbb{E}[\log q(y)]$ um that's not going to
give you something that's a true
probability distribution
right it's not going to be a thing that
sums to 1 right where probabilities all

together sum to 1.

what it's going to do is give you this
negative you know a bunch of negative
numbers

right so the softmax function does is it
normalizes that which just means that it
takes the kind of relative values
you know of each of those things and
turns it into

a probability distribution that sums to
one so

so you end up with that bolded pi symbol
is just a a probability distribution
that's assigned

a π that assigns a probability to each
possible policy

and a policy by the way is just a
sequence of actions

right so one policy might be like turn
left turn right and another policy might
be turn right turn left

um and so it's just saying that um
that that minus f minus g thing will
just be turned into a probability
distribution

and that probability distribution over
different policies is what the
uh agent will sample from right so if
the

if π says this policy is 0.8 and this
other policy is 0.2

then the agent will be a lot more likely
to choose the policy that's 0.8
yeah maybe go to the last caveat slide
and then

i have a few other questions where we'll
just try to like have a question

simple answer question simple answer and

then we'll move into the part two screen
share

okay so the two caveats are just a there
are two extra components of the pl mdp
that is namely the e

vector which you can see is kind of
pointing to

or there's that e block that's pointing
to policies

and then there's our gamma as well
that's also pointing to policies

and so e is how generally speaking how
one models habit formation or one way of
modeling habit formation

and so i'm not going to talk about that

now i just kind of want to flag it

and then when we come up things come up
in learning you won't be surprised

there's now there's extra term in the
policies and get what gamma does gamma
is essentially weights the contribution

it's a

of g to policy selection and so the idea
is roughly speaking is that

g is essentially a you have your prior
distribution over policies

which doesn't take in f at all and then
you

take you have your posterior
distribution which takes in f

and then you look at the difference
between those and

if there's a large difference between
those roughly speaking

then gamma will go down it will
essentially down weight

the contribution of expected free energy
to your posterior over policies

and this is kind of linked to has some
really this is how we model dopamine
phasic dopamine spiking for example
i'm not going to get into any more
details there because i think it's
actually much easier to just have like a
fully worked example but i did just want
to flag all that stuff
okay so i think that was my last so
that's like this tightening
of strategy while things are effective
but not overfitting but a tightening
while things are working
and then during a mismatch a period of
high uncertainty or ambiguity
there's actually a movement towards more
exploratory behavior
um okay let me try to go through a
couple
of these questions so we'll just try to
have the like sort of clippable
one one further thing i just wanted to
clarify about that though is that
so so gamma thing and really you should
think about that as kind of like the
precision estimate for expected free
energy so so all it's doing
is just saying hey if things were super
different
than what i thought they were before
policy-wise
then i mean in some cases a little more
complex than that but but basically
i can't trust my expected free energy
estimate as much
right right so i should down weight i
should sort of down weight
how much me expected free energy

contributes to what i choose to do
because it's not as reliable right so
it's just saying how much should my
habits
and my current observations affect what
i do versus

what i expected ahead of time uh how
rewarding they were going to be
right so if the clock has always been
accurate and then you look at you're
like wow i didn't know it was 1pm
you're like i believe the clock but if
you know that the clock is inaccurate
then you think i thought it was noon i
looked at it it said one but you know
who knows with this clock these days
so that's about the confidence in the
observation okay

so policy selection two quick questions
on that

the first one is what is the time scale
of policy selection

and does this model assume any
particular temporal scale

um so i mean temporal temporal scales
are kind of just whatever you want them
to be

right so so like you know in the context
of a task

right you might just say here's time
point one which is where

you know a participant gets some
observation right and here's time point
two which is where

they choose an action right in which
case that would be a two

time step trial um

and i mean it doesn't really matter

right i mean it might the stimulus might
be on for five seconds
or it might be on for one second in the
first you know
time step in the trial um and maybe
their decision takes
half a second right so the time step two
could be half a second
or it could be like a minute right i
mean like so it's
time steps are kind of whatever you
define them to be relative to a task
in another session when we get to
hierarchical models you will have
um higher levels that operate on slower
time scales
than the lower levels but that's a kind
of more complex thing that we'll have to
cover later
okay another policy related question uh
is can you comment on different forms of
policies
such as plans from $\tau = 0$ to
big time t so from the beginning till
 t and then τ equals little t to big t
belief action policies research followed
by belief action policies
and then how is this related to
different constructs such as working
memory
and habits so maybe just first part of
the question was
how are different kinds of policy
estimation
undertaken and the second part is how is
that related to maybe different
constructs we have
with regular vocabulary like working

memory or habits

um do you want me to take that curse or
do you want to um yeah i'm happy to take
it i suppose

uh so you can have shallow policies or
you can have deep policies

where deep policies are just kind of
time point whatever time point one
to t and you're just summing over
basically all of

you basically it's the path integral
formally speaking just over
expected free energy and that is updated
at each time point

um or you can have shallow policies
whereas there's one time step
uh how does this relate to like things
like tree search and all that stuff
um well it depends so

i s i'm assuming that that question
means that you're kind of
familiar with like bellman formulations
and that kind of thing

so expected free energy is bellman
optimal for one-step policies
and it's sub it's not bellman optimal
for deep policies

but there are sophisticated what's
called a sophisticated inference scheme
which is is this correct ryan that's
formally equivalent to uh backward
induction

yeah yeah so yeah sophisticated version
of active inference

which is more sort of explicitly like a
deep tree search

um is is equivalent to backward
induction which is film and optimal yeah

exactly um and so there are forms of
this that do relate to kind of research
and in regard to working memory
uh depends on how you want to set up the
task so there are like models
of uh say working memory where you have
two level models where your model
working memory is kind of the thing that
has
stable maintenance uh you can then or
you can model it with kind of one level
models
uh it just it just depends on how you
want to set things up generally speaking
i think of
working memory as being functionally
defined there are lots of definitions i
won't put one out there right now but
like
something to do with maintenance right
and then it kind of just depends on what
you're trying to model about working
memory what type of model is appropriate
yeah i mean one one additional thing to
say is just really kind of basic is that
working memory is gonna
come down to what your transition
matrices look like
right so if you're if you're put into
state one
and you have a really strong belief in
your transition matrices that you're
gonna if you start in state one you're
gonna stay in state one
right then that more or less amounts to
that state remaining active
over over several time steps and if you
learn that at time step one

and then at time step four believing in
that same believing you're in that same
state
tells you what action to do then that's
a type of working memory so uh thomas
parr has a really awesome
prefrontal computation as active
inference where he kind of thinks
or use thinks about transition precision
in terms
of excitatory recurrent connections in
lateral pfc essentially which have been
linked to things like maintenance
cool one last quick take before we go to
ryan sharing your screen is related to
modeling
a given behavioral task what are the
criteria
for setting the factors where the model
will be embedded
so how do we just sort of operationalize
the kinds of things that we take into
account in this model
i believe is what the question is asking
um i mean there's in a lot of tasks
there's probably not one unique way
right i mean that's kind of the where
the the kind of creativity
problem solving aspect of this comes in
is because you know you kind of have to
figure out
right what what uh
factorization structure what generative
model structure whether it's factorized
or not
right or what different sets of ways to
set up a model
that can um generate

uh behavior that you would see in a task
right i mean i'll give you
i'll give you an example right when we
build this task
build this task model together in my
part here um
but you know general in a lot of cases
you can make something factorized or you
can make it not factorized
um and you know sometimes you can set
actions up so that uh it has to do with
um probabilities in the b matrix and
your transition beliefs and in other
cases you can set it up so it's in the
likelihood
in the a matrix um so it's not there
isn't
my point is there isn't one unique
solution always for
for what generative model to use and a
lot of times you might
try out multiple generative models and
then do model comparison to figure out
which one's best
yeah all right maybe while you're
setting up your
screen i'm going to ask one more
question
somebody wrote i'm still not exactly
clear on why you would want to maximize
information and minimize the difference
between your desired outcomes and your
expectation
what advantage is afforded by maximizing
information
if it doesn't enhance the likelihood of
receiving the desired outcome
uh well it won't it won't in that case

like typically what will happen is
is if there's so the agent i mean it
depends a little bit on the
exact task setup but what will usually
happen is that
if the agent knows what to do to get the
reward
then that just means that that cost term
or that kind of reward probability term
will just be the one that dominates like
the value of that will dominate the
thing will just
select reward automatically the
information seeking
will typically only have a high weight
if it's the case that the agent doesn't
know what to do yet
to get to get the reward you know that
reminds me a lot
it reminds me a lot of the game theory
uh strategy which is like tit for tat
but then you start out playing nice
it's like default to being cooperative
but then have a strategy
and so it's sort of like a meta approach
which is if it's
if it's in the room if you're winning
the game of go or chess
at whatever training level then what is
there to do
your strategy is working but then when
there isn't a victory observation
then it entails an exploratory search
that percolates through higher and
higher abstractions
of learning in the system so i hope that
we can
make that a little bit more tangible

with this session right here
so we have 40 minutes left and again
this is just part one of multiple
so let's go for you know 20 to 30
minutes followed by
10 to 20 minutes of wrap up and final
questions from the chat and a couple
other ones i have stored up
but thanks um and take it away ryan
okay so i want to just ask real quick
can you guys see my like mouse if i like
point to things
yes okay okay because i i i need to
point to things when i'm presenting this
stuff
so okay so so just kind of again to get
to
scope here i mean the end goal of the
tutorial is for people to be able to
actually do research with us with these
things
right so so the end goal here is to
learn how to build task models
for empirical studies and the basic idea
is that you're going to have
participants perform a task
and then what you want to do is find the
parameter values in the model
um say like values for precision or
values for prior
expectations or values for
habit distributions right that e thing
that um
i'm not gonna probably i won't use e
very in this
example but but just things like that
right where
different things in a model could take

on different values
and based on those values the agent will
act differently
it'll make different choices and so what
you want to do is
take the behavior of an actual
participant and then
find the values for those parameters
there is that
they'll best reproduce their behavior
um and uh so
and then once you have those parameters
that best reproduce
um a given person's behavior and you do
that for all the participants
then you can just use those values those
parameter values
as individual difference measures you
might say like if one person has a
higher precision
uh a higher a matrix precision
than another person you know does that
predict something about
for instance how well they're going to
respond to a treatment like in
computational psychiatry or
can that tell you something about um you
know some other cognitive function
or i mean you know et cetera et cetera
um
and um so the goal again is to
get people to that point where they can
do that um
and um like i said before this is the
kind of thing that people have been
doing for a long time
in reinforcement learning um but it's
new

it's a novel sort of not very common
thing yet to do an active inference
because not enough people know how to do
it

um and so you know i can give you uh
examples here if i can get my what the
uh it's um

i don't know why this is not letting me
advance

[Music]

um okay there

um so you know and these tasks and vary
you know in another session we'll

um you know walk you through how to
build like a perceptual task that's like
widely used for like

eeg research um as well as number
imaging research more broadly but
called like an oddball task which is
primarily a perception task

and there are also sorts of inferential
like prospective decision making tasks
that you can use that don't involve
learning

and then there are tasks that you know
that

use reinforcement learning or explore
exploit dynamics

um and then you can also use it with
neural imaging

right you could say like trial by trial

where are the prediction errors

you know in the brain what parts of the
brain look like they're doing the free
energy minimization

um you know trial by trial um so there's
things like that that you can do
why okay

and so sorry what just that that really reminds me of just to pull back a level making measurements of a biological system like gene expression or neuroimaging and then saying what gene is upregulated in people who have this condition or what brain region is activated in this condition and we're taking that beyond the simply descriptive beyond the just measure and do a t-test and we're thinking about these observations these behavioral phenotypes as being emitted often across tasks potentially or across modalities emitted by a generative model that has these higher level parameters that shape the way that the person or the organism responds to stimuli under a certain specific framing of a time and task dependent model just to sort of pull back a level and give that but continue right so so i just want to say that you know there are this is kind of a new a new thing to do not very many people have done it um you know my lab has kind of been uh trying to kind of get this approach out there more um and so i mean here's some example papers that just do each of the things that i mentioned right so

this one on the right here so recent
plus computational biology paper
is using active inference primarily just
for a perceptual model
um and with that one we were just
looking at um actually uh
inter-receptive processes right so for
instance
um every time people feel a heartbeat
and then indicate by pushing a button
that they feel their heartbeat
you know who who does better than that
uh than others
and why right and some people might have
stronger prior expectations that they
are or not going to feel a heartbeat and
some people instead might just have
different levels of sensory precision
right like they treat the signal coming
up from their body as more or less
precise
right so that's like one example um
this paper uh in drug and alcohol
dependence was a learning
an action or a explore exploit
and reinforcement learning task that was
applied to people in substance use
and then this one in the lower left is
greater decision uncertainty
paper um is an example of
uh purely just sort of planning like
decision making inferential model
without learning
um and then philip schwartenbeck's paper
up here on the right is an example of
using active inference to
look for the neural correlates of
particular uh

model parameters um you know so for instance with the with the drug and alcohol dependence paper and just use this simple exploit tasks where people had three options they could push button one two or three and then you could either a green or red ball would kind of fall and green meant they won and red meant they lost and initially they didn't know you know what the probability was of winning or losing for each winning or losing for each button um so the first few choices they had to kind of explore right to figure out which one's giving the most greens i mean eventually they become confident that one of them is best and they kind of stick to that one and i won't go through it but you know you can specify a particular version of the graphical models that uh chris showed that generate appropriate behavior like participants do on this task and again i won't really go through this but this is what the likelihood or a matrix would look like that they would where over time they would learn these a zero values to basically say if i choose a slot machine one two or three what's the probability that that's going

to generate uh

this row the observation for this row

which is a win or the observation for

this row which is a loss

and then the c vector here the one that

encodes reward

just says whatever this cr value is

which would be a positive number

just says i prefer winning versus losing

by some amount

and then this is the equation for

learning and then this is just showing

um this is just showing the expected

free energy equation

um and again this isn't the example i'm

going to work through

but what you could do i'll skip that is

you could find

that people with substance use showed

lower

action precision in other words the

behavior was a little more random

they showed higher learning rates for

wins

and lower and less slower learning rates

for losses compared to healthy people

right so you can tell this interesting

story about hey like if people with

substance use

learn more slowly whenever a bad thing

happens when they take the drug but they

learn more

you know more quickly every time they

feel good right after taking a drug

then um their behavior is going to have

a much harder time

stop stopping uh taking taking a a

drug um despite the fact that it has

negative consequences
um you know and so you can fit uh these
are just
group estimates just showing in a
bayesian way that
learning rate for losses was uh
different um in people with uh substance
use than healthy people
and uh the example for just uh
planning and decision making without
learning this was a
an approach avoidance conflict task
basically people just had to decide
where to move this little
avatar dude on uh along this little
runway
line and the closer they were to like
the sun
and uh and a bar that had lots of red
that mean meant they were going to see a
nice happy image and they were going to
win some points the more red and the
uh rectangle thing the more points they
would win
and the rainy cloud meant they were
going to see like a really
terrible picture and hear like a
terrible sound
um so in some cases they would need to
say okay
i'm willing to go through seeing this
really negative
uh aversive thing to win a lot of points
or not right so in this model it's just
trying to say
you know they start in the middle for
example you know
how close do they decide to get to one

side or the other which
um controls the probability of getting
what's on one side or the other
so it's just planning where to move
right there's no learning because they
already know exactly
what the rewards and punishments are um
and so in that case again i won't go
through this but you can specify a
generative model for this
um in a similar way you can run
simulations about what we people will do
under different um parameter values this
beta thing is the um
expected precision um for expected free
energy precision
um and ec here is just how much they
like the reward like the points relative
to how much they dislike the negative
images
and we just called that emotion conflict
um
that's just another way of presenting
the generative model i won't go through
that
but here again you can show for example
that
people with depression and people with
substance use
showed uh less emotion conflict than
healthy people
and they showed greater uh decision
uncertainty than healthy people they
showed they had
higher uh a lower expected free energy
precision
which corresponds to higher beta values
um so that's just another example

um i think one thing that uh from the presenter view you can't see questions but now that i'm not in presenter view i can see that daniel had his hand up oh sorry oh okay just just yes this is great stuff i just want to highlight a few pieces here so with that same task you can imagine a purely descriptive model and a paper being written on something like people who have x diagnosis are more likely to approach when there's conflict or something so that would be a purely descriptive finding based upon the same exact data now what this tutorial is about is using the exact same behavioral data within and across participants and then modeling underlying parameters that have a very specific graphical layout and relatedness so here we're talking about a t-test basically being used on a summary statistic so they're not just different in the descriptive outcomes like this group was more likely to gamble on red this is like saying this group had a higher hidden state variable that we're going to attach to what we're going to call like gambling propensity now that relates to a question from the chat which is um and a really important question how are the neurobiological

interpretation of the agent's generative
model parameter
estimates done with respect to the
parameter model so how do we go from
a summary statistic or a graphical model
estimate to a neurobiological
interpretation let alone intervention
or anything like that but just mainly
the interpretation side
um so so we have a whole a whole section
in the tutorial on the neural process
theory that's associated with active
inference
um that's something that we're going to
uh that was we plan to talk about
in another session um probably in part
two
um so i'm probably going to hold off on
describing that in detail right now
but the the kind of quick version um
is that uh the the process the neural
process theory just makes certain
assumptions
about how uh neurons can be connected up
to do this kind of thing
um you know so for example um there's a
you can have different rates at which
your beliefs change
rates with which the with your
distribution over states
more or less precise um and
you know the the neural process theory
says that the for instance the the sorts
of erps
that you would measure right the changes
and the changes in uh the magnitude of
change in the uh
neural activation that you would see

would correspond to the rate of change
in in those beliefs essentially you can
think about it as how quickly prediction
error is being minimized
as well um whereas the
the updates to this beta thing um trial
by trial
um are uh modeled as the
as the phasic dopamine spikes that are
essentially updating
um how much the expected free energy
controls uh action slow you know policy
selection
um so there are there is just there's
just a neural process theory that
proposes one way in which the uh the
variational message passing that
otherwise these models
how that could be implemented in early
and therefore what sorts of signals
are you know you ought to measure in the
brain if that process theory was correct
but it's it's really important to
recognize that there are several
different
possible process theories that you could
come up with right so
there's there's many different ways you
you could
connect a bunch of neurons up together
to implement these sorts of models
so there's like a lot of different
levels of um
study here right there's what generative
model best describes task behavior
and then there's given that model what
of a bunch of different message passing
algorithms

is the one that is being used
and then there's the question about
given that it's method this message
passing algorithm
you know there's a bunch of different
possible ways the brain could implement
that so which one is the right one
right so there's many different levels
of of you know empirical questions that
that you can ask
at each of these different levels of
description yep
what i'll say to that is that the
relationship between the free energy
principle
and active inference again as a process
theory
it means something uh within the
philosophy of science literature
so check out the alias 2018 interview
with
carl fristen myself and marchand fortre
or the recent paper of mel andrews
because basically the process theory
makes specific
falsifiable testable hypotheses so
that's the kind of thing you could
actually then say dopamine use this kind
of tool
and you should expect to see this and
that's why this is a little bit more
towards the data
focused empirical end so continue right
yeah but like i said i mean we'll go
into it you know task models will make
specific predictions about
what kind of erps you know you would
measure in uh

eeg uh you know during certain tasks or
same thing with like fmri responses
and and so but anyway we'll we'll get
into that in another session
um so just uh you know so this is
another example again i'll walk you
through these really briefly
but that's what that's another
generative model we built for this
heartbeat
perception task and what we found here
was that
in a certain condition a certain high
arousal condition
healthy people showed higher precision
uh higher intraceptive precision
whereas a bunch of different clinical
groups didn't show any change in
precision
so again it's just showing differences
here in the way the brain treats signals
coming up from the body
when in a high arousal state um or
really a state where they're having to
hold their breath for a long period of
time but um
and then that's you know you can also
estimate prior expectations those didn't
differ by group
um then finally this is an example of
doing uh using active inference for
neural imaging that
philip schortenbeck did and here he was
looking at
the the beta gamma updates the expected
free energy precision updates
and he showed that trial by trial those
did those updates did correlate

with these midbrain region with this
midbrain region um
that is where a chunk of that region is
is um
where a bunch of dopamine neurons are um
so sort of consistent with this kind of
idea that
phasic dopamine responses are the ones
encoding these changes and expected free
energy precision
um anyway so those are just a bunch of
examples of the way that this kind of
thing has been used
and i should point out that this is all
just within the last couple of years
this is very recent
um so the the task that um i'm going to
walk you through how to actually build
it
um and we'll see how much time we
actually have to go through this
um is a pretty simple task where
you just the participant just starts in
the start state
and initially they don't know anything
about they have to choose one of two
of these slot machines the one on the
left or the one on the right
um and they don't know you know i have
no idea to start with on trial one
what which one is more likely to uh to
give a reward
and if you're in the the left better
context then that means that
choosing the left slot machine will win
80 percent of the time
and if you're in the the right better
context

then choosing the right slot machine
will give you the reward
um eighty percent of the time um and so
crucially
if on the first time step they just
choose a slot machine
then uh they could they'll win four
dollars if they're right
um but the other things that'd be kind
of like the reward seeking thing you
know they want as much money as possible
um but what they can also do is they can
first
ask for this hint if they choose to ask
for the hint
it will tell them what context they're
in it will tell them whether it's left
better or write better
and then they can choose one of the two
slot machines and again they'll win 80
of the time
if they choose the right one based on
the hint but crucially
if they take the hint first then they'll
only win two dollars if they get it
right
instead of four so in other words taking
the hint is costly
um so so you can think of choosing one
of the slot machines right away as being
this kind of
reward seeking thing where it's also
risk seeking
because you don't know ahead of time
which one's the right one
or you can do this information seeking
thing where you choose the hint first
even though that'll get you less money

but you'll be more confident which one's right

so it's set up specifically to have this information gain component by choosing the hint

and the reward seeking component by choosing one of the two slot machines so the question is how would you actually in practice build a model of this kind of task

now the first thing you know so so there's going to be some basic steps to do this

one is to define whatever your initial state priors are

and that's going to be that d thing um

and just as a you know we'll go into this more with the learning

but big d if you use both big d and little d

big d is kind of like the true generative process um you know the real thing out in the world that are generating the observations

um and little d if you use it is a is what's in the generative model so the the agent's beliefs

right and those two can be so big d and little d can be different if big d and little d are the same that means

that the agent's beliefs are accurate um and so

if you want the agent to learn or to have different prior expectations than what the true ones are

then you have to use this little d thing um

same thing here we have to define the

the likelihood or the state outcome
mapping

and that's going to be our big a and our
little a if we want it to

learn and have different beliefs than
the true ones

we have to define the preferences over
outcomes right so that's going to be the
 c thing then we have to divide to find
the possible transitions

or actions um and so you know if there's
just one transition matrix for a factor
that just means the agent has no control
over

what it does but if there's a state
factor

that can have multiple possible
transitions

where each transition is like like an
action

then uh the agent can choose policies
that correspond to different
transition sequences um you know so for
instance

well i'll just show you um but then but
then for

v here the last thing you have to define
is policies which is

this thing v in the code and that's
going to be specifying different
sequences of b matrices different
sequences of

possible transitions that could happen
over the course of the trial and that
the agent could choose

one of those transition sequences um
yeah really really quickly for each of
those then if

if there's uh you know if i'm talking
about a different outcome modality i
would have
i would have to have one of those
parameters for each outcome modality is
that correct
um you would need to have so there's
going to be
one uh a matrix
for each outcome modality and that's
just going to say
what uh what outcomes are going to most
likely to be generated
um given each combination of of states
um so given the value of each of each uh
state factor um and and these are things
that are
that come up in experimental design
right so like the probability that a
given machine would dispense
a winning ticket or that's something i
should be thinking about as the
empirical person designing my experiment
right
yeah exactly so so i'll show you
um here but right so typically you can
do it other ways but typically
this likelihood thing is what would
define the um the reward probabilities
right so you'd say given that i'm in the
state of
uh you know having chosen the left
uh slot machine um that will generate
reward the reward observation with 0.8
and the not reward observation as point
as 0.2
for example such as saying the reward
probability is eighty percent

um and you know the agent would
if you want them to learn those word
probabilities then you'd specify one of
these little a things and then you'd
have it
learn it um over repeated observations
um
it's such a critical and and really
fascinating point that
reward isn't absent from the model
through the preference vector c
it's baked into how policy is calculated
and so it's like the agent is pursuing
precision
all things being equal through natural
selection you know
unsuccessful models are just not going
to exist and models that
don't see themselves performing behavior
are not going to be active for long
so under a model of a successful
preference then there's just a
convergence towards that with whatever
affordances are at hand
so it's just a really interesting way to
see how re
uh reinforcement learning situations
um can be adapted so
the atms some of the some of it gets a
little bit more into the kind of like
free energy philosophy stuff but but
i mean yes like the assumption is that
uh people inherit preferences that keep
them alive
um but uh but in in kind of like an
empirical task context you just use the
 c vector to just
define what counts as the reward um

you know it could be it could be
anything right it could be winning money
it could be winning
points even if it doesn't give you money
right it could be like
seeing a positive image you know
anything like that you just define
what the preference distribution is um
and then usually you parameterize it so
you
fit that as a parameter to see
essentially how much does the person
like
you know winning two dollars for example
um
and i'll show you that in a second um
but uh but yeah
so so you're gonna need one c uh
 c matrix uh for each so you need one a
matrix and one c matrix for each output
modality
um and you're gonna need one d vector
and at least one v matrix for each state
factor
um as the to answer your uh the original
question
um so so the first thing here
right is you the first so the way that
i decided to build the model for this
task is
the first hidden state factor is the
context
right so am i in the state where the
left one is better
or am i in the state where the right one
is better
and i started that out specifying this
little d here thing

um and the brackets just say the number
of the state factor so this is d
for state factor priors over states for
state factor one
and there's two different possible
states left better or write better
and i just specified these as two really
small numbers
which just says it's an even probability
that the it's going to be the left one
or the right one is better
but these are really small and so i'm
not confident at all
about you know whether that distribution
is right
um and i'll explain more about that when
we get to the learning
um because this is this will be soft
maxed
um during inference so this will become
0.5
during inference but when you learn you
sort of
build up the numbers here so they become
bigger numbers
which means the agent learns to be more
confident in what the distribution is
that it believes
but then i can also specify big d right
which is the generative process
and i just put this as one and zero
which means that the true context is
that the left one's better
um so that's what i would use to set up
the priors for uh state factor one
um state factor two here um if you can
see
at the bottom um so

hold on can you guys you guys see this
or can you guys see the bottom because
uh
for in mine i can see like my bottom the
like
we only see one line of code yeah the d2
bind yeah i'm just asking do you see
like this like search down here and all
that stuff
it's perfect it's perfect okay there now
it's gone okay
so so then d2 is the second
hidden state factor and that's the one
that corresponds to your choices
right so this is a one here which means
it always starts out in the
start states the first column here is
the start state
and then it can either transition into
the hint state right which is the second
one here i just define that
um and then the third one is the choose
the left
slot machine state and the fourth one is
choose the right slot machine state
so this is just says at time one my
prior
is with 100 certainty that i'm going to
start in the start take
of the task um so that's actually pretty
simple
right so then the next thing you're
going to do is you're going to have to
specify the likelihood so the a matrix
and i'm not going to show you the
complete one yet until we go into the
code i'm just going to show you the
parts that matter

um but so for the second state factor
so a2 so uh for the second outcome
modality
so the likelihood mapping for a two um
is going to be this thing where
the first row corresponds to null which
is just it hasn't won
or lost yet the second row is the
observation that it lost
and the third row is the observation
that it won
and each of the columns here correspond
to
the state factor values so the left
column here corresponds to
the d1 you know this thing you know
uh so the 0.25 here for left better for
the left better state
and the right one here is the one that
corresponds to the light better state
which maps onto column two here so the
way that you would read this matrix is
that if you were in the right state
right better
state then you would
win with this probability
and you would lose with this probability
um or if you were in the left one then
you would lose with that probability and
win with that probability
um the columns of that soft maxed
are those are those distributions yeah
the columns the columns are shaft maxed
yeah again if you're using i mean to get
a little technical if you're using
if you're using little a so these are
dirichlet distributions
then it will build up counts right so

these numbers could become like
you know like 50 25 8 you know whatever
but those that encode the confidence in
those distributions but yeah they'll get
soft max for inference
um so then
so then you'd have to also specify the c
matrix
for that state factor or i mean that
outcome modality sorry so c_2
is going to correspond to a_2 basically
it's for the sec they're both for the
second outcome modality
um and here again rows are going to be
observations
but here the uh the rows correspond to
time points
in the trial um so basically this is
just saying
that um at time 0 or at time 1
i have no preference for anything at
time
2 i have a negative preference for
losing
and i have a positive preference for
winning and this r_s thing is just a
parameter right so we can say this l_a
thing so loss of version
is one so it doesn't want to lose with
negative one
right um and for r_s we say r_s equals
four
which is you could think of as the four
dollars right
and then so if it ch if it observes the
win
at time two then it's gonna prefer that
with a value of four

and if it's gets it at time three then
it's gonna be rs divided by two which
means at time three if it gets it it's
only gonna win two dollars
right or whatever the relative values
are for the person right you could fit
rs for a given person um to see how much
they
prefer four dollars over two for example
um
and um these also um in the code are
soft maxed and then
uh logged um so they become they become
long probabilities
um um so that's just saying
how much you dislike losing and how much
you want to win
and again you can set those as
parameters
and then for state outcome
for outcome modality one which is the
getting the hint right you can have no
hint
you can have the hint that the left
machine is better and the hint that the
right machine is better right so those
are both observations
and then you can say uh with this pha
thing you can say basically how
how informative is the hint right like
if i observe the machine
left hint does that tell me with
certainty that the left one's better or
does it just tell me with some
probability that the left one's better
um and here you could just set this as
uh being
you know like you could just set uh this

is like ones and zeroes
um to just to just tell you uh
basically that would just say uh if p_h
was one then that would just say that
the uh
the thing is a hundred percent uh
accurate
um in telling you which one's better if
you observe one hit versus another
um and um
and so then the next thing right is you
have to consider the possible actions
or sequences of actions that are going
to correspond to policies
and so here the way to think about it is
so you're starting out
in state one first day factor two right
the like action
state factor and you can either
go from the start state immediately to
choosing the right
machine or you can go immediately from
the start to choosing the left machine
or you can choose to take the hint and
then go to the left one or to the right
one or take the hint and then go to the
left one
so those are like the the those
sequences of actions are the policies
that are gonna
matter right they're gonna encode what
the agent can do
um and so to do that you have to set up
different transition matrices different
b matrices
that encode each of these actions right
transitioning from one
to four transitioning from one to three

transitioning from one to two
transitioning from two to four and two
to three um
and so that's going to correspond to
these setting up these b matrices
which um here so b_1
right which is the state factor for uh
the um
which one is which slot machine is
better um
this is just going to be an identity
matrix which basically just says
that the left better context is constant
across the trial
you know it's not as if from time point
one to time point two
the context is gonna change or something
like that right this just says the
belief is
and it is true that each trial has a
stable identity as being the left better
one or the right better one
um but for state factor two so
 b_2 here that's where you want the agent
to have different possible actions
right so b_2 colon colon one
right so the third dimension being the
different uh different possible action
um the first one would be like this
which basically just says so columns
are states at time t
and rows are states at time t plus one
um so this just says i can start in any
state
so in any column and choose to move to
state one this one says i can start in
any column
and move to state two so taking the hint

this one says i can start in any column
so any state and move to
state three which is using the left
machine then same thing here for
changing the
to moving to the right machine so there
are four actions one two three and four
they correspond to four different third
dimensions
in the b matrix for state factor two
um and so then given those
um given those
we specify v which are the policies
um and here the third dimension
corresponds to
which state factor so for this one for
state factor one
there are no actions right it's just
this one v matrix and its identity
matrix
and here the rows are the action and the
columns are
sorry the rows are the time point and
the columns are the uh
the action or the policy sorry
um and so the row here just means action
moving from time one to time two
and the uh um and this second row
means the action from time to time three
um and so that
is useless right for state factor one
because there's just only one action
whereas for state factor two we have all
the different possible action sequences
right so policy one would just be if the
thing just decided to stay in the start
state the whole time right like to just
not do the trial um the other one would

be action two which is take the hint
and then and then take action three
which is move to the
left slot machine or take the hint
it to the third column take the hint and
then move to four which is the action of
choosing the right slot machine
um and then here this is just kind of a
little bit of a trick
but you can choose to at time from one
time to one to time two you can
immediately choose the left slot machine
and then move back to the start state so
three and then one
or same thing choose the right slot
machine and then again move back to
state one
and the only reason for that move back
to state one thing is is that um
if you let the thing stay in state three
and state four
then it's as if it won four dollars and
then won two dollars after that
so you have to kind of move it back out
of the state where it would win
um so so then
you know then you've built everything
really um
so you and you throw all of it in this
little mdp
structure um so i didn't mention this
but you have to specify
t which is the number of time points in
the trial so in this case it would be
uh three um so start take the hint
choose left
start take the hinge who's right et
cetera so there's three time points

um v is just v which is what we defined
um we'll ignore u for now but that's
what you would use if you wanted to
specify one step policies instead but
the thing doesn't look ahead at all um
you know and then a b c and d are just
the different matrices we specified and
we want it to learn

d um so we specify the little d um
and then these other things are just
different parameters that you can set
um so ϵ is the as a learning rate um
 α is kind of like an action
precision so it's basically an inverse
temperature parameter it controls how
random someone's
choices are given the policy that they
choose um

β is the expected free energy
precision that we talked about
and um these other two parameters i
won't really go into detail but they
just have to do with specifying
um the time constant so basically how
quickly evidence accumulates
after an observation and this τ thing
is um

basically it controls it controls
certain assumptions about
what ought to happen if you would make a
new observation each time with respect
to what sorts of
neural responses you would get um
and this is all explained in the code
but i think that's a

it's a perfect pause point so that we
can close within the hour
and leave people with excitement for

part two so

what can they look forward to in part

two and beyond in a minute

yeah so i mean i would just say that you

know once once you have this set up

then the next steps are to run this

structure through

this vbx underscore tutorial script and

that will actually run the model

and you know simulate the behavior the

neural responses and then we'll show you

how to

use um plotting scripts

to display and show how that behavior

show the behavior that uh was the

outcome of the simulation

and then we'll show you how the the kind

of structure the mdp structure works

um and then actually show you some

simulation results and how they work and

things like that

well and i actually probably walk you

through some of the actual code

um that was an amazing session

ryan christopher max thanks so much for

coming on

everybody who was watching live and in

replay also

very appreciated so please leave a

comment

if you have a question or feedback for

the authors or for anyone else

and stay in touch with us because we're

going to be making this a multi-part

series where we're going to be going

deeper into the technical aspects

and also highlighting use cases hearing

about people who are just learning

programming learning to apply it people
who are experts in other fields so
whatever your perspective is you're in
the right spot to be learning
so thanks again all of you for coming on
and we will see you another time